Technology and Health Care 17 (2009) 253–267
DOI 10.3233/THC-2009-0545
IOS Press

An online lifestyle diary with a persuasive computer assistant providing feedback on self-management

Olivier A. Blanson Henkemans[a,b,*], Paul J.M. van der Boog[c], Jasper Lindenberg[b], Charles A.P.G. van der Mast[a], Mark A. Neerincx[a,b] and Bertie J.H.M. Zwetsloot-Schonk[c]
[a]*Delft University of Technology, Delft, The Netherlands*
[b]*Dutch Institute of Applied Science TNO, Soesterberg, The Netherlands*
[c]*Leiden University Medical Center, Leiden, The Netherlands*

Abstract. In accordance with the global trend, in The Netherlands approximately 45% of the population is overweight. Existing studies show that patient self-management can reduce these figures, but medical non-adherence is a persistent problem. eHealth can potentially increase adherence to self-management. Consequently, we designed a persuasive computer assistant and evaluated its influence on self-management, i.e., the use of an online lifestyle diary called DieetInzicht.nl. The assistant is represented by an animated iCat, which shows different facial expressions and provides cooperative feedback following principles from the motivational interviewing method. We conducted a randomized controlled trial with 118 overweight people over a period of four weeks and studied the difference between diary use with and without computer assistant feedback. Results show that the computer assistant contributed to filling in the diary more frequently, reduced the decline in motivation to perform self-management, lowered the (reported) BMI, and improved the ease of use. Furthermore, diary use increased knowledge of maintaining a healthy lifestyle. Finally, personal characteristics, i.e., locus of control, vocabulary, computer experience, age, gender, education level and initial BMI, explained the variance in the diary use and its outcome. Of the 118 participants 35 filled in the closing survey, covering motivation, BMI, lifestyle knowledge and ease of use, which implies that the findings based on these results are mainly representative for motivated participants. In general, this study shows that the Dieetinzicht eHealth service, including a personal computer assistant, is likely to support motivated overweight people and lifestyle related diseases to get a better insight in and adhere to their self-management.

Keywords: eHealth services, overweight, online lifestyle diary, persuasive computer assistant, self-management, e-coaching

1. Introduction

Globally, one billion people are overweight [35]. In the Netherlands, it impacts approximately 45% of the population. Moreover, obesity affects approximately 9% of the population [40]. Besides causing personal distress [8,21], being overweight may lead to an increase in the number of patients with lifestyle related and chronic diseases, such as diabetes and cardiovascular diseases [25,41]. Studies have indicated that more patient engagement in the care process could reduce these figures, e.g. [27,31]. Consequently, to address this situation, there is a gradual shift from a passive relationship between patients and the

*Address for correspondence: O.A. Blanson Henkemans, Delft University of Technology, Mekelweg 4, 2826 CD, Delft, The Netherlands. Tel.: +31 152786331; Fax: +31 152787141; E-mail: O.A.BlansonHenkemans@TUDelft.n.

health care system towards a more independent, self-determining position wherein the emphasis lies on self-management.

Self-management consists of activities undertaken by individuals, families, and communities with the intention of preventing disease, limiting illness, and restoring health [27]. It aims at educating and increasing the patient's intrinsic motivation, which in turn can lead to stimulation of medical adherence [31]. Examples of self-management activities are maintaining a healthy diet [17,25] and performing physical activities [10,23]. Strikingly, although people are well informed of the impact of being overweight on their life and from the benefit of self-management, treatment non-adherence is a persistent problem [15,20].

eHealth can potentially stimulate adherence to self-management [1,26,34,42]. Remote care for people at home through Information and Communication Technology (ICT) has multiple benefits, such as lowering costs [18], aging in place [36], motivating the patient [19] developing health literacy [28,33], and assessing personal characteristics that influence specific care needs [43]. Illustratively, the online support can be provided by informative websites, which help the user to search for relevant medical information independently and develop health literacy. Also, it can facilitate personal data management, which can lead to a better overview of the user's own behavior.

In one study [5], patients with rheumatoid arthritis received internet-mediated physical activity intervention with personal counseling. Home-based patients had access to the website, called Cybertrainer, for one year. Results show that this intervention was more effective with respect to the total amount of physical activity than a similar program without additional facilities.

Other studies looked at the influence of personalized computer assistance on patient self-management and eHealth, concerning feedback [6,7] and representation [29]. In regard to feedback, the assistant applied a cooperative feedback style, i.e., it has a coaching character, explains and educates the patient, and expects high participation of the user. Or it applied a directive feedback style, i.e., it has an instructing character with brief reporting and expects low participation of the user. Results showed that the cooperative assistant was more effective and satisfactory, whereas the directive assistant was more efficient. In regard to representation, the use of animated personas and robots can stimulate and improve the quality of the interaction [24]. For example, research on the use of a robot for the support of diabetes treatment showed that both children and older adults preferred interacting with a social intelligent animated iCat over a simple text interface [29,30]. These studies took place in smart home laboratories with healthy adults.

Although the studies described above have shown the value of eHealth services and personal computer assistants, there is little prolonged research on how the use of a personal computer assistant can improve diet and exercise behavior of people who are overweight. Moreover, in medical research on human participants, consideration of the well being of the subject should take precedent over the interest of science [13,46]. Consequently, the goal of our study was to evaluate the support of the persuasive computer assistants of the use of an online lifestyle diary.

This article discusses the implementation of a persuasive computer assistant in the online lifestyle diary DieetInzicht.nl. Subsequently, it discusses the influence of the Computer Assistant on the self-care process, i.e., the use of the diary, and outcomes, i.e., motivation, weight and health literacy. This will takes place according to a randomized control trial with adults, who are overweight, at home.

2. Online lifestyle diary with persuasive computer assistant

Following the discussed research on self-management, eHealth services, and personalized computer assistant, we designed a persuasive computer assistant and integrated it in an existing PHP and MySQL based lifestyle diary called DieetInzicht (www.dieetinzicht.nl). The goal of the diary is help users obtaining better insights on how to maintain a healthy lifestyle without external manipulation. It provides insight in how to maintain a healthy lifestyle by offering objective information in relation to diet and physical activity. The users create a personal account with username and password. With the username and password, they log in and access a personal diet and exercise diary, report, and product pages. On the personal page, participants manage their personal data, i.e., weight and password. Moreover, they set their personal self-management goals, regarding diary use, diet and exercise. Examples of goals are, respectively, "Fill in diary four times a week", "Eat a healthy amount of fat", and "Exercise lightly on a daily basis". As illustrated in Fig. 1, in the diet page they keep a diary of the amount of products they consumed. They specify the product, category, e.g., vegetable or meat, quantity and time. In the activity page, they keep a diary of the exercise performed. They specify the time, category, e.g., intense or lightly intense, and time of day. In the report page, they receive a report on nutrition components consumed, including, calories, fats, proteins, and carbohydrates, and number of calories burned during their activities. Finally, on the product page, they browse for products and accompanying nutrition facts.

In the lifestyle diary, we implemented a persuasive computer assistant on the DieetInzicht website. The assistant offers support by monitoring the diary and providing cooperative feedback, which is coaching and aims at patient engagement. Participants are free to pick goals, which are important to them. In turn, the assistant indicates if the participant is fulfilling his or her goal or not. Also, it will give suggestions on how to achieve goals in the future by referring to relevant pages on the website of the Voedingscentrum, the Dutch association for nutrition.

When giving feedback, the computer assistant follows principles of the motivational interviewing method [32]. Research shows that the application of motivational interviewing, which focuses on social functioning, discussing problems and giving feedback in the form of advice and direction, has proven to be an effective therapeutic approach [38]. The computer assistant does not have the full range of motivational interviewing abilities of a human caregiver. However, we applied some principles of this method in our computer assistance that previously was well-appreciated by the users, such as the HealthBuddy, a telemonitoring system for patients with heart failure, and the SuperAssist iCat, a robotic assistant for diabetics [29,30,44]. In accordance with the motivational interviewing principles, the computer assistants:

– *Expresses empathy:* The assistant is mindful that the participant might have other priorities ("You were not successful in achieving your goal for today. Maybe you have been busy.");
– *Is cheering and complimenting:* The assistant congratulates the participants when goals are achieved ("Well done! You are successful in achieving today's goal.");
– *Explores discrepancies between life style goal and current lifestyle:* To give a good overview of the situation, the assistant indicates what goals are not realized and suggests how this could be improved ("You were not successful in achieving your goal for today. Too bad. But do not get discouraged and try again tomorrow.");
– *Supports self-efficacy and optimism:* The assistant stays optimistic about achieving goals and underlines that it is acceptable to make errors and the participant has the ability to realize goals ("You were not successful in achieving your goal for today. This is not terrible. It will go with small steps, so keep trying the coming days.").

Both the personal lifestyle goals and the feedback were evaluated and improved where necessary by an independent dietician from Leiden University Medical Center (LUMC).

To illustrate the functioning of the computer assistant, Table 1 lists the diary components according to the control system model [12]. In brief, the PHP website is connected to a MySQL database, which

Fig. 1. Diet page of online lifestyle diary with persuasive computer assistant.

contains the users' personal details, personal goals, and entries in food and activity diary. The database also contains descriptions and dependencies of all personal goals (4 for each category, i.e., diary, diet and exercise) and a list with related textual and graphical feedback. When accessed, the website runs the MySQL queries to check if the user is achieving his or her goals or not. Depending if the query outcome is positive or negative, the website shows the related textual feedback accompanying iCat animations.

As illustrated in Fig. 2, the assistant is represented by an animated iCat, which shows three facial expressions: neutral, happy and sad. Research has shown these facial expressions can contribute to the understanding of the computer assistant [4]. Moreover, it can enrich the experience of empathy by the assistant [29,30]. In our study, when opening the website, the assistant looks neutral, i.e., blinks its eyes, occasionally looks around. When the user addresses its goals, the assistant will look happy or sad depending on achieving the goals or not. The iCat is a research platform created by Phillips for studying human-robot interaction. It is a plastic yellow cat with a face and a body, which expresses emotions by

Table 1
Control system model [11] and diary components

Control system model components	Description	Diary components	Computer assistant activity
Input function	Perception of current situation	MySQL database with users and their diary entries	Assess participant's food and exercise diary entries
Reference value	Standard, set point, or goal	MySQL database with users' lifestyle goals	Retrieve participant's diary, diet, and exercise goals
Comparator	Device that makes comparisons between input and references value	Computer Assistant model of comparison between goals and diary entries based on MySQL queries	Based on diary entries, test if goals of each category is fulfilled or not.
Output function	Error corrective behavior	MySQL database with Computer Assistant feedback based on comparison	Based on the test results, select feedback text array, pick relevant texts from feedback list list, and select the happy or sad iCat animation.

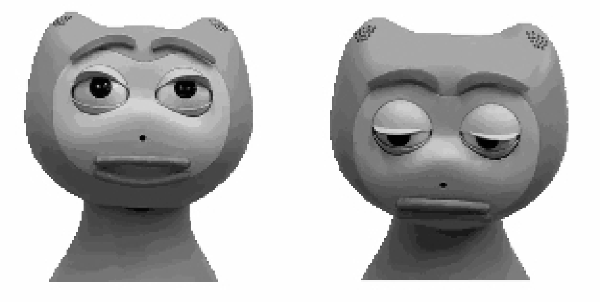

Fig. 2. Examples of happy and sad facial expressions of the persuasive computer assistant in accordance with feedback.

moving its lips, eyebrows, eyes, eyelids, head and body [9].

During our study, our main research question reads: Can a persuasive computer assistant increase the adherence to an online lifestyle diary for overweight adults? Based on previous studies on eHealth services, we expect that participants receiving persuasive feedback from the computer assistant will use the diary more adequately, better fulfill lifestyle goals, be more motivated [2,7], and develop more self-managing knowledge [33]. Also, we expect that active use of the diary will lead to improved self-management outcomes [19], such as optimizing the Body Mass Index (BMI). Finally, we expect that personal characteristics, i.e., demographics, cognitive abilities and personality traits, will influence the use of the diary [14]. To validate our hypotheses, we conducted a randomized control trial. Overweight adults used an online diary from their home computer for a period of four weeks. With this method we expect to obtain valid empiric findings for the benefits of eHealth services [3].

3. Methods

Over a period of three months, we recruited 191 Dutch overweight adults, who had no previous experience with the DieetInzicht website and were not under periodic treatment with a specialist at a hospital. People are considered healthy when they have a BMI between 18.5 and 25 kg/m^2 and are considered overweight when they have a BMI between 25 and 30 kg/m^2. A BMI above 30 kg/m^2 is

Table 2
Frequency table of male and female participants with and without computer assistant

	Recruited	Maintained diary >5 days (male/female)	Completed all surveys (male/female)
Computer assistant	97	65 (15/50)	18 (3/15)
No computer assistant	94	53 (6/47)	17 (3/14)
Total	191	118 (21/97)	35 (6/29)

Table 3
Surveys issued and number of participants that completed the specific surveys

Survey	Pre-study (Day 01)	Study (Day 01–28)	Exit survey (Day 28)
Demographics	118		
Computer Experience	118		
Locus of Control Scale	118		
Vocabulary	118		
Lifestyle knowledge	118		35
Motivation	118	50 (on Day 14)	35
BMI	118		35
Daily diet and physical activities		118	
Online diary evaluation			35

considered obese. They were recruited through the Dutch Research Institute for Applied Science (TNO) participant database with overweight people and an advertorial in an online Dutch national newspaper, which ran for one week. Of the 191 participants, 118 subjects filled their diary for at least 5 days and were included in the study. As listed in Table 2, the final sample consisted of 21 male and 97 female participants, between the age of 21 and 65 ($M = 43.24$, $SD = 11.55$). The average BMI was 27.90 ($SD = 2.42$) and 65 participants had a computer assistant. Furthermore, 35 participants completed all surveys. Participation was voluntarily. There was no communicative stimulus or financial compensation, but among the participants that fulfilled the study, 5 prizes, worth 150 Euro each, were administered, chosen by lottery. The study protocol was approved by the Dutch Medical Ethical Testing Commission (METOPP).

Following the randomized controlled trial, the diary autonomously assigned the participants, double blinded, to the study group or control group. The study group set lifestyle goals, filled in the diary and received feedback from the computer assistant. The control group performed the same activities, but did not have a computer assistant providing feedback. To assess the influence of the computer assistant, we measured diary use, concerning the number of days people used the diary and the number of days between the first and last day of use, and the average number of products and physical activities entered per day. Furthermore, we studied adherence to lifestyle goals, motivation, health literacy, BMI, and usability. Finally, we studied the influence of personal characteristics on the use of the diary.

As listed in Table 3, participants started the study with completing an online survey concerning demographics, BMI and computer experience. Also, they completed a survey on self-management motivation [45]. This survey addresses the expectation to be able to use the diary and maintain a healthy lifestyle. Furthermore, they completed the locus of control scale test, which measures if people have a predominantly internal locus of control, i.e., attribute events to their own control, or external locus of control, i.e., attribute events to external circumstances [37]. Finally, they completed a vocabulary test (Dutch translation of Shipley Institute of Living Scale [39]), and lifestyle knowledge test, which was validated by the LUMC dietician.

The first three days after signing in, participants were free to explore the website and adjust the goals. Then, participants received an email instructing to choose three lifestyle goals, one each for the diary, diet

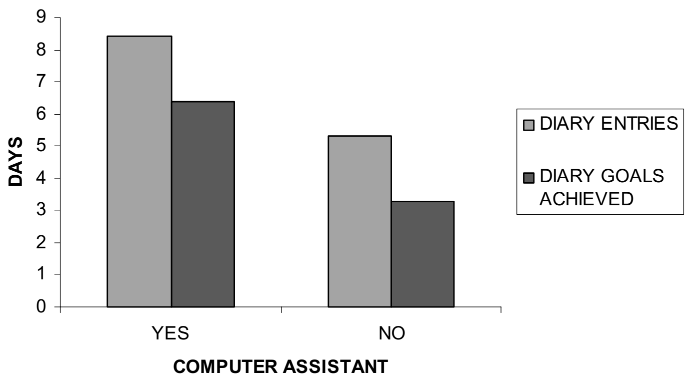

Fig. 3. Mean frequency of diary entries ($M = 8.43$, $SD = 9.11$ and $M = 5.30$, $SD = 7.12$), ($t(116) = 3.04$, $p < 0.05$) and diary goals ($M = 6.38$, $SD = 8.96$ and $M = 3.30$, $SD = 5.88$), $t(116) = 2.15$, $p < 0.05$) achieved, in days, with and without computer assistant ($N = 118$).

and physical activity, for the remainder of the study. On day 14, participants received an email referring to the survey on self-management motivation. On day 28, participants received an email referring to the closing online survey. It assessed self-management motivation, lifestyle knowledge, Body Mass Index, and evaluation of the diary, concerning the diary's contribution to a healthy lifestyle and its usability. The usability survey consisted of 7 point Likert scale questions about the contribution of the diary to maintaining a healthy lifestyle, ease of use, relevance, trust, and educational value. Finally, the participants received a debriefing document.

To evaluate the influence of the computer assistant on the self-management process, we performed a T-Test for independent variables. To evaluate the influence of the computer assistant on the self-management outcomes, concerning health literacy, motivation, and BMI differences across the four weeks period, we performed a repeated measure ANOVA. To determine whether personal characteristics explain variance in the use of the diary and its outcome, we conducted a multiple regression analysis with six predictors whereby we assume linearity, random sampling, no perfect collinearity between the independent variables, and independence. The predictors were age, BMI, education level, computer experience, vocabulary, and locus of control.

4. Results

In regard to diary use, as illustrated in Fig. 3, participants who had the computer assistant, filled in their diary with higher frequency. Of the 28 days, they filled in their diary more often ($M = 8.43$, $SD = 9.11$), than when having no computer assistant ($M = 5.30$, $SD = 7.12$), $t(116) = 3.04$, $p < 0.05$. Moreover, participants with computer assistant filled in their diary over a longer range of days, with a maximum of 28, ($M = 9.70$, $SD = 9.44$) then people without a computer assistant ($M = 6.51$, $SD = 7.73$), $t(116) = 1.98$, $p < 0.05$.

Following the frequency of diary use, also as illustrated in Fig. 3, participants with a computer assistant achieved their daily diary goals more often, i.e., more days ($M = 6.38$, $SD = 8.96$), than participants without a computer assistant ($M = 3.30$, $SD = 5.88$), $t(116) = 2.15$, $p < 0.05$. Results did not show a significant influence of the computer assistant on adherence to diet, i.e., the number of days they achieved their diet goals ($M = 0.86$, $SD = 2.48$) and exercise, i.e., the number of days they achieved their diet goals ($M = 0.63$, $SD = 3.59$).

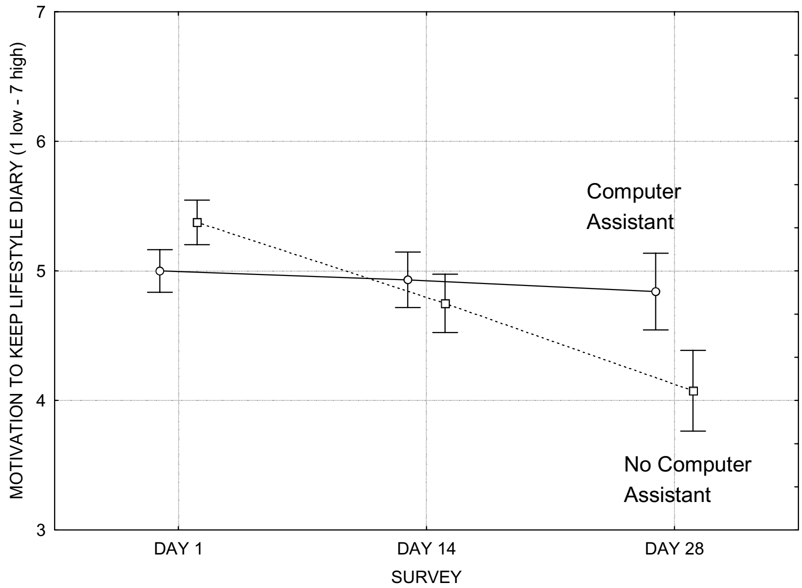

Fig. 4. Mean score for motivation to keep a diary (1 low – 7 high), at the beginning ($M = 5.38$, $SD = 0.59$ and $M = 5.00$, $SD = 0.53$), halfway through ($M = 4.75$, $SD = 0.66$ and $M = 4.93$, $SD = 0.68$), and end of the study, with and without computer assistant ($M = 4.84$, $SD = 0.96$ and $M = 4.07$, $SD = 0.94$) ($N = 35$), with an interaction effect, $H(1,32) = 4.47$, $p < 0.05$.

At the beginning, halfway through and the end of the study, we surveyed the participants' self-management motivation, concerning motivation to keep a diary and to maintain a healthy lifestyle. Participants' score ranged from 1, not motivated at all, to 7, very motivated. The number of participants that filled in the survey halfway through and at the end decreased and we are confronted with a considerable amount of missing values. Consequently, we conducted a Kruskal-Wallis ANOVA analysis with the 35 participants (18 with and 17 without computer assistant) who completed all three surveys. As displayed in Fig. 4, participants' self-reported motivation to fill in the diary declined over the period of four weeks, $t(65) = 4.75$), $p < 0.001$. However, the participants with computer assistant declined less than the participant without a computer assistant, $H(1,32) = 4.47$, $p < 0.05$. As displayed in Fig. 5, participants' self-reported motivation to perform self-management declined over the period of four weeks, $t(63) = 1.91$, $p < 0.05$. In contrast, the participants with computer assistant increased, $H(1,32) = 5.56$, $p < 0.05$.

At the beginning and end of the study, participants completed surveys on knowledge of maintaining a healthy lifestyle. As illustrated in Fig. 6, the 35 participants who filled in both surveys, scored better at the end ($M = 8.84$, $SD = 1.11$), than at the beginning of the study ($M = 7.63$, $SD = 1.58$), $F(1,30) = 50.81$, $p < 0.001$. There was no difference between participants' score with and without computer assistant.

Considering the 35 participants who entered their BMI at the beginning and end of the study, people with a computer assistant scored an average BMI of 28.39 ($SD = 2.32$) at the beginning of the study and an average of 27.78 ($SD = 2.32$) at the end. People without a computer assistant scored, on average, a BMI of 28.23 ($SD = 2.36$) at the beginning of the study and 28.16 ($SD = 2.49$) at the end. As displayed in Fig. 7, people working with the diary lowered their BMI significantly. Moreover, people with the computer assistant decreased their BMI even stronger, $F(1,28) = 13.84$, $p < 0.001$.

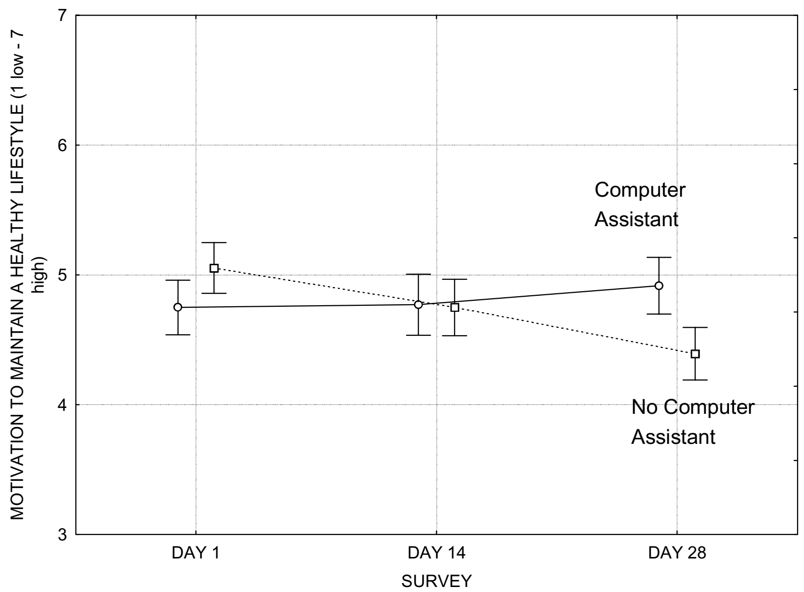

Fig. 5. Mean score for motivation to maintain healthy lifestyle (1 low – 7 high) at the beginning ($M = 4.66$, $SD = 0.87$ and $M = 5.03$, $SD = 0.84$), halfway through ($M = 4.73$, $SD = 1.13$ and $M = 4.68$, $SD = 1.02$), and end of the study, with and without computer assistant ($M = 4.82$, $SD = 0.78$ and $M = 4.15$, $SD = 0.86$) ($N = 35$), with an interaction effect, $H(1,32) = 5.56$, $p < 0.05$.

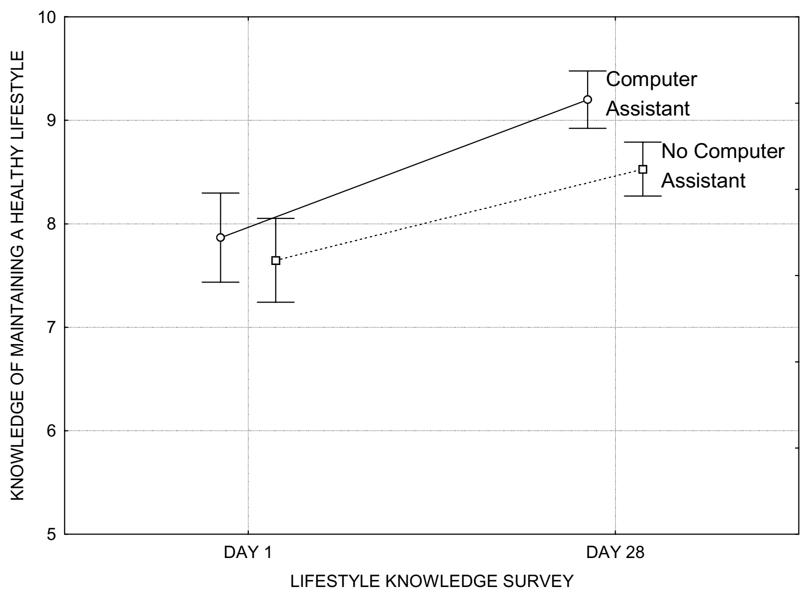

Fig. 6. Mean score for knowledge of maintaining a healthy lifestyle at the beginning and end of the study, with and without computer assistant ($N = 35$), $F(1,30) = 50.81$, $p < 0.001$.

At the end of the study, participants evaluated the diary's usability. As listed in Table 4, on a scale from 1 (low) through 7 (high), 35 participants, 18 with and 17 without computer assistant, rated its contribution to maintaining a healthy lifestyle, ease of use, relevance, trust and educational value. In

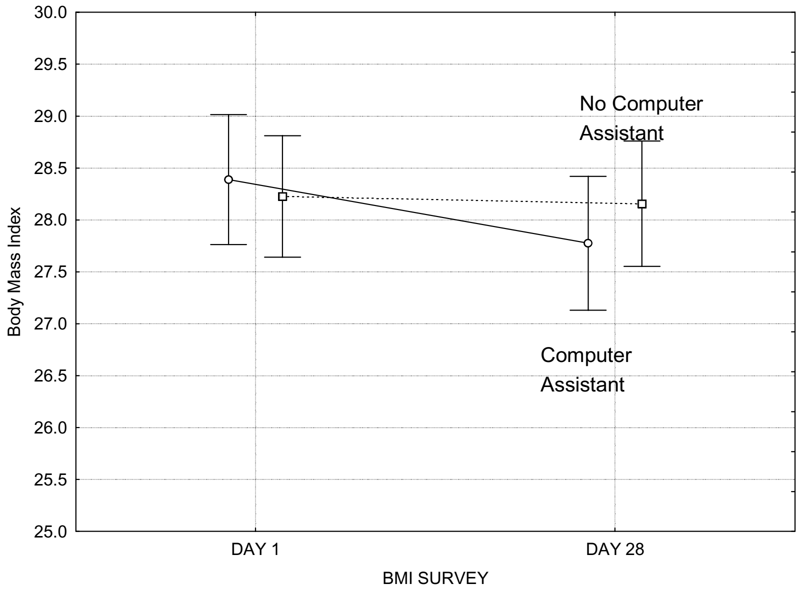

Fig. 7. Mean Body Mass Index at the beginning (M = 28.39, SD = 2.32 and M = 28.23, SD = 2.36) and end of the study (M = 27.78, SD = 2.32 and M = 28.16, SD = 2.49), with and without computer assistant (N = 35), with an interaction effect, $F(1,28) = 13.84$, $p < 0.001$.

Table 4
Evaluation of diary usability (1 low – 7 high) (N = 35)

Usability aspect	Mean	Standard deviation
Contribution to maintaining a healthy lifestyle	4.29	1.18
Easy of use	4.43	1.38
Relevance	4.83	1.10
Trust	4.29	0.99
Educational value	4.29	1.27

addition, we analyzed the influence of the computer assistant on these aspects. As displayed in Fig. 8, results of a Mann-Whitney test, show that participants with computer assistant appraised the diary as easier to use (M = 4.89, SD = 1.32), then participants without a computer assistant (M = 3.94, SD = 1.30), $Z(34) = 2.27$, $p < 0.05$.

As listed in Table 5, results of the multiple regression analysis explain the variance in diary use and outcomes by personal characteristics. Concerning diary use, Locus of Control, BMI, vocabulary, computer experience, and gender explain 23% of the variance in diary use completeness, $F(5,111) = 6.52$, $p < 0.001$. Concerning diary use outcomes, BMI, computer experience, age, gender explained 37% of the variance in self-reported motivation to maintain a healthy lifestyle halfway through the study, $F(3,21) = 4.95$, $p < 0.05$, and 21% at the end of the study, $F(2,22) = 3.75$, $p < 0.05$.

5. Discussion

In a randomized controlled trial, we evaluated the influence of a persuasive computer assistant on the adherence to an online lifestyle diary for self-management. In the diary, overweight adults created a personal account, set lifestyle goals, kept track of dietary and physical activities, and received a report

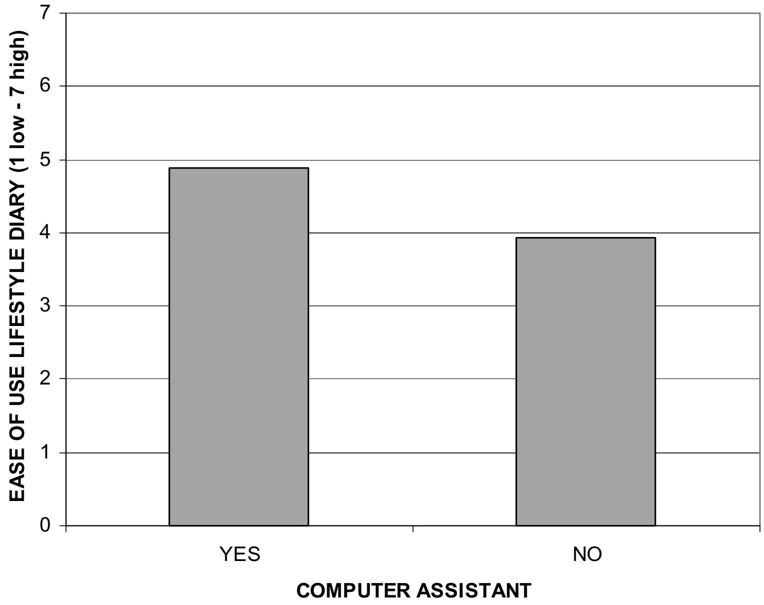

Fig. 8. Mean rating of diary's usability aspect "Ease of use" (1 low – 7 high), with (M = 4.89, SD = 1.32) and without computer assistant (M = 3.94, SD = 1.30), (N = 35), Z(34) = 2.27, $p < 0.05$.

Table 5
Multiple regression analysis of variance in online lifestyle diary completeness, motivation and usability explained by personal characteristics

Variable	Explained variance by predictor variables	R^2 (%)	Regression equation
Average diary entries in per day	Locus of control (LOC)	10	Completeness = −16.92 + 0.31*LOC + 0.24*BMI + 0.20*VOC − 0.14*CE + 0.12*G
	BMI	5	
	Vocabulary (VOC)	4	
	Computer Experience (CE)	2	
	Gender(G)	2	
	Total	23	
Motivation to maintain a healthy lifestyle, beginning of study	BMI	11	Motivation = 5.90 − 0.40*BMI − 0.44*VOC + 0.44*CE
	Vocabulary (VOC)	7	
	Computer Experience (CE)	15	
	Total	33	
Motivation to maintain a healthy lifestyle, halfway through the study	Computer Experience (CE)	21	Motivation = −39.11 − 0.45*CE + 0.25*Age − 0.23*EL
	Age	5	
	Education Level (EL)	5	
	Total	29	
Motivation to maintain a healthy lifestyle, end of study	Computer Experience (CE)	16	Motivation = 4.65 − 0.40*CE + 0.19*EL
	Education Level (EL)	4	
	Total	20	

of consumed nutritional products and burned calories. The assistant is represented by an animated iCat, which shows different facial expressions and provides cooperative feedback on the diary entries following the motivational interviewing concept. We studied the influence of the assistant on the use of the diary, achieving lifestyle goals, self-management motivation, health literacy, BMI, and usability.

Results of our study show that adding the persuasive computer assistant contributed to using the diary and achieving diary goals more frequently, reduced the decline in motivation to perform self-management, lowered the participants' BMI, and increased the diary's ease of use. Furthermore, the use of the diary increased the participants' knowledge of maintaining a healthy lifestyle. These findings are in accordance with earlier presumed benefits of eHealth [34]. Overall, the diary was evaluated as usable; moreover, the computer assistance contributed positively to the diary's ease of use. Finally, personal characteristics explained the variance in completeness of diary use and self-reported ability to perform self-management.

These findings have implications for designing eHealth services. The use of a computer assistant, represented by an avatar, which provides cooperative feedback, applies motivational interviewing and accompanies the feedback with facial expression, can contribute to self-management adherence. It can support overweight people to keep a diary more adequately and fulfill personal lifestyle goals, stay motivated over time, lower their BMI, and increase knowledge on how to maintain a healthy diet and perform exercise. Consequently, in accordance with existing research, our findings show that people adhering to self-care, which is stimulated by our assistant, can improve their health condition and maintain it in the long run [31]. Lastly, when designing eHealth, we must consider the influence of personal characteristics, such as age, education level, computer experience, locus of control and vocabulary on the use of technology. More specifically, participants who had a high internal locus of control, and scored high on vocabulary, computer experience, entered the diary more completely and younger participants, who scored higher on computer experience and education level, were more motivated to maintain a healthy lifestyle.

When interpreting these results, we must consider three important issues. First, of the 191 recruited participants, only 118 fulfilled our requirement of using the diary for at least 5 days. This has implications for our findings. Mainly, we see that, in line with the literature, e.g. [22], it is difficult to stimulate people to maintain a healthy lifestyle through eHealth over a longer period of time. This is especially the case when, like in our study, the experiment leader had no personal contact with the participants and people learn about the online lifestyle diary through an advertisement or mailing and create an account independently and anonymously. However, the initial goal of the diary is to obtain insights in how to maintain a health lifestyle. Apparently, after using the diary a number of days, the users feel they achieved this goal and thus feel less compelled to continue entering their food and activities in the diary. Consequently, it could be beneficial to add features to the diary which makes it more attractive to use it continuously. Examples of features are a recall function, which reminds the user of the diary, an information library with interesting news on diet and exercise, and a community service that facilitates communication with peers. Also, it is difficult to guarantee the accuracy of diary entries and survey of demographics and physical measures, such as BMI. Consequently, to assess the exact influence of the use of the diary on health condition requires ongoing clinical testing.

Second, relatively few men participated in the study. Existing research suggests two possible causes. A study shows that women are more socially-orientated and men more individually-orientated [16]. As a result, women would be more willing to participate in our voluntary study than men. In addition, in the past decennia, women increased their dissatisfaction with their body shape. Also, women are more likely to judge themselves overweight than men [11]. In accordance with these theories, men would be less likely to see the importance of the online lifestyle diary.

Third, of the 118 eligible participants, only 35 completed all surveys, which assess influence of the computer assistant on self-management motivation, health literacy, BMI, and usability. This entails a bias in the closing survey outcomes, i.e., the computer assistant leads to better diary use outcomes amongst the most motivated participants, which makes the effect of the assistant on less motivated people arguable. Still, there are arguments that the findings are repetitive for the whole population. Namely, the computer assistant increased adherence to the diary amongst all eligible participants and, in the group that completed all surveys, the number of participants with and without computer assistant was equally divided.

In conclusion, the persuasive computer assistant contributes to the adhering to the usable online lifestyle diary by overweight adults. Participants with assistant used the diary more frequently, achieved diary goals more regularly. Moreover, motivated participants with assistant stayed more motivated, and lowered their BMI. Finally, use of the diary contributed to lifestyle knowledge. Consequently, this eHealth service is likely to support people with lifestyle related and chronic diseases, such as diabetes and cardiovascular diseases, to adhere to their self-management. This includes maintaining a healthy diet and regularly performing physical activities. Finally, personal characteristics, i.e., locus of control, vocabulary, computer experience, age, gender, education level and initial BMI, influence the use of the online diary and its outcomes and need to be taken into consideration when designing eHealth services.

Acknowledgments

Research was supported by SenterNovem IOP-MMI, of the Dutch Ministry of Economic Affairs, supporting the SuperAssist project. Also, we want to thank Ires Schulze, dietician at LUMC, for her contribution.

References

[1] D.K. Ahern, J.M. Kreslake and J.M. Phalen. What is eHealth: Perspectives on the Evolution of eHealth Research, *Journal of Medical Internet Research* **8**(1) (2006), e4.
[2] L.L. Alpay, R.I. Overberg and J.H.M. Zwetsloot-Schonk. Empowering Citizens in Assessing Health Related Websites: A Driving Force for Healthcare Governance, *International Journal of Healthcare Technology and Management* (*IJHTM*), *Special issue: "Governing ICT in healthcare and governing healthcare with ICT"* **8**(1/2) (2007), 141–160.
[3] K. Ball, D.B. Berch, K.F. Helmers, J.B. Jobe, M.D. Leveck, M. Marsiske, J.N. Morris, G.W. Rebok, D.M. Smith, S.L. Tennstedt, F.W. Unverzagt and S.L. Willis, for the ACTIVE Study Group. Effects of Cognitive Training Interventions With Older Adults. A Randomized Controlled Trial, *Journal of the American Medical Association* **288** (2002), 2271–2281.
[4] P. Barkhuysen, E.J. Krahmer and M. Swerts, Problem Detection in Human-Machine Interaction Based of Facial Expressions of Users, *Speech Communication* **45** (2005), 343–395.
[5] H.M. van den Berg, H.K Ronday, A.J. Peeters, E.M. Voogt-van der Harst, M. Munneke, F.C. Breedveld and T.P.M. Vliet Vlieland, Engagement and satisfaction with an Internet-based physical activity intervention in patients with rheumatoid arthritis, *Rheumatology* **46**(3) (2007), 545–552.
[6] O.A Blanson Henkemans, M.A. Neerincx, J. Lindenberg and C.A.P.G. van der Mast, SuperAssist: Supervision of Patient Self-Care and Medical Adherence, in: *Proceedings of IEA2006 Congress*, R.N. Pikaar, E.A.P. Koningsveld and P.K.M. Settels, eds, Maastricht: Elsevier, July 2006.
[7] O.A. Blanson Henkemans, W.A. Rogers, A.D. Fisk, M.A. Neerincx, J. Lindenberg and C.A.P.G. van der Mast, Usability of an Adaptive Computer Assistant that Improves Self-Care and Health Literacy of Older Adults, *Methods of Information in Medicine* **47** (2007), 82–88.
[8] B. Blaine, J. Rodman and J. Newman. Weight Loss Treatment and Psychological Well-being, *Journal of Health Psychology* **12**(1) (2007), 66–82.
[9] A. Breemen, X. Yan and B. Meerbeek, iCat: an animated user-interface robot with personality, *Autonomous Agents and Multiagent Systems*, AAMAS'05. ACM Press, New York, NY, USA, 2005, 143–144.
[10] A.G. Brooks, N.F. Butte, W.M. R and, J.-M. Flatt and B. Caballero, Chronicle of the Institute of Medicine Physical Activity Recommendation: How a Physical Activity Recommendation Came to Be Among Dietary Recommendations, *American Journal of Clinical Nutrition* **79**(5) (2004), 921S–930S.
[11] T.F. Cash, J.A. Morrow, J.I. Hrabosky and A.A. Perry, How has body image changed? A cross sectional investigation of college women and men from 1983 to 2001, *Journal of Consulting and Clinical Psychology* **72** (2004), 1081–1090.
[12] C.S. Carver and M.F. Scheier. Control Processes and Self-Organization as Complementary Principles Underlying Behavior, *Personality and Social Psychology Review* **6** (2002), 304–315.

[13] C. Coyle, G. Doherty, M. Matthews and J. Sharry, Computer in talk-based mental health interventions, *Interacting with Computers* **19** (2006), 545–562.
[14] S.J. Czaja, N. Charness, A.D. Fisk, C. Hertzog, S.N. Nair, W.A. Rogers and J. Sharit, Factors Predicting the Use of Technology: Findings From the Center for Research and Education on Aging and Technology Enhancement (CREATE), *Psychology and Aging* **21**(2) (2006), 333–352.
[15] S. van Dulmen, E. Sluijs, L. van Dijk, D.T.D. de Ridder, E.R. Heerdink and J. Bensing, Patient Adherence to Medical Treatment: A Review of Reviews, *Health Services Research* **7** (2007), 55.
[16] C. Eckel and P. Grossman, Are Women Less Selfish Than Men? Evidence from Dictator Games, *The Economic Journal* **108**(448) (1998), 726–735.
[17] European Food Safety Authority (EFSA). Development of Food-Based Dietary Guidelines, *Summary Report EFSA Scientific Colloquium* 5, 21–22 March Parma, Italy, 2006.
[18] G. Eysenbach, What is e-Health? *Journal of Medical Internet Research* **3**(2) (2001), e20.
[19] B.J. Fogg, *Persuasive Technology: Using Computers to Change What We Think and Do*, San Francisco, CA: Morgan Kaufman Publishers, 2003.
[20] R.B. Haynes, X. Yao, A. Degani, S. Kripalani, A. Garg and H.P. McDonald, Interventions to Enhance Medication Adherence, *Cochrane Database System Reviews* **19**(4) (2005), CD000011.
[21] E. Jeurissen and M. van Spanje, *Rondom Dik: Zin en Onzin Over Zwaarlijvigheid* (About Overweight: Sense and Nonsense About Being Overweight). Amsterdam: Ambo, 2001.
[22] B. Kaplan and S. Litewka, Ethical Challenges of Telemedicine and Telehealth, *Cambridge Quarterly of Healthcare Ethics* **17** (2008), 401–416.
[23] H.C.G. Kemper, W.T.M. Ooijendijk and M. Stiggelbout, Concensus Over de Nederlandse Norm voor Gezond Bewegen (Concensus on the Dutch Norm for Healthy Exercise), *Tijdschrift Sociale Gezondheidszorg* **78** (2000), 180–183.
[24] E.J. Krahmer, Z.S. Ruttkay, M. Swerts and W. Wesselink, Pitch, eyebrows and the perception of focus, in: *Proceedings of Speech Prosody,* B. Bel and I. Marlien, eds, Aix-en-Provence, 2002, pp. 443–446.
[25] C.F. Kreijl, A.G.A.C. Knaap, M.C.M Busch, A.H. Havelaar, P.G.N. Kramers, D. Kromhout, F.X.R. van Leeuwen, H.M.J.A. van Leent-Loenen, M.C. Ocke and H. Verkley, eds, Ons Eten Gemeten. Gezonde Voeding en Veilig Voedsel in Nederland (Our Food Measured. Healthy Nutrition and Safe Food in the Netherlands). *RIVM Rapport 270555007, Volksgezondheid Toekomst Verkenning RIVM*. Bilthoven, the Nederlands. Retrieved February, 2008, from [http://nl.sitestat.com/rivm/rivm-nl/s?270555007&ns_type=pdf&ns_url=[http://www.rivm.nl/bibliotheek/rapporten/270555007.pdf].
[26] C. Larizza, R. Bellazzi, M. Stefanelli, P. Ferrari, P. De Cata, C. Gazzaruso, P. Fratino, G. D'Annunzio, E. Hernando and E.J. Gomez, The M2DM Project: the experience of two Italian clinical sites with clinical evaluation of a multi-access service for the management of diabetes mellitus patients, *Methods of Information in Medicine* **45**(1) (2007), 79–84.
[27] H. Leventhal, E. Halm, C. Horowitz, E. Leventhal and G. Ozakinci, Living with Chronic Illness: A Contextualized Self-Regulation Approach, in: *The SAGE Handbook of Health Psychology*, S. Sutton, A. Baum and M. Johnston, eds, 2004, pp. 159–194.
[28] K.D. Lippa, H.A. Klein and V.L. Shalin, Everyday Expertise: Cognitive Demands in Diabetes Self-Management, *Human Factors* **50**(1) (2008), 112–120.
[29] R. Looije, F. Cnossen and M.A. Neerincx, Incorporating guidelines for health assistance into a socially intelligent robot, *Proceedings of 15th IEEE Ro-Man*, 2006, Sep 6–8, University of Hertfordshire, Hatfield, UK, 2006.
[30] R. Looije, M.A. Neerincx and F. Cnossen, Persuasive robotic assistant for health self-management of older adults: Design and evaluation of social behaviors. Submitted to *Special issue on Human-computer interaction for medicine and health care* (*HCI4MED*) *International Journal of Human-Computer Studies* (*IJHCS*).
[31] S. Maes and P. Karoly, Self-regulation Assessment and Intervention in Physical Health and Illness: A Review, *Applied Psychology An International Review* **54**(2) (2005), 267–299(33).
[32] W.R. Miller and S. Rollnick, *Motivational Interviewing: Preparing People to Change Addictive Behaviour*, New York: Guilford Press, 1991.
[33] D. Morrow, D. Clark, W. Tu, J. Wu, J. Young, M. Weiner, D. Steinley and M.D. Murray, Correlates of Health Literacy in Older Adults with Chronic Heart Failure. *Gerontologist* **46** (2006), 669–676.
[34] J. de Nooijer, A. Oenema, G. Kloenk, H. Brug, H. de Vries and N. de Vries, *Bevordering van Gezond Gedrag via Interent. Nu en In de Toekomst* (Promotion of Healthy Behavior Through Internet: Now and In the Future). ZonMW, Retrieved February, 2008, from http://www.gvo.unimaas.nl/voeding/PDF/bggi.pdf.
[35] P. Puska, C. Nishida and D. Porter for World Health Organization (2003). *Obesity and Overweight*. Retrieved on February, 2008, from http://www.who.int/hpr/NPH/docs/gs_obesity.pdf.
[36] W.A. Rogers and E.D. Mynatt, How Can Technology Contribute to the Quality of Life of Older Adults? in: *The Technology of Humanity: Can Technology Contribute to the Quality of Life?* M.E. Mitchell, ed., Chicago, IL: Illinois Institute of Technology, 2003, pp. 22–30.

[37] J. Rotter, Generalized Expectancies for Internal Versus External Control of Reinforcements, *Psychological Monographs* **80** (1966), 609.

[38] E.K. Santa Ana, S. Martino, S.A. Ball, C. Nich, T.L. Frankforter and K.M. Carroll, What is usual about "treatment-as-usual"? Data from two multisite effectiveness trials, *Journal of Substance Abuse Treatment* **35**(4) (2008), 369–379.

[39] W.C. Shipley, *Shipley Institute of Living Scale*, Los Angeles: Western Psychological Services, 1986.

[40] Statistics Netherlands (CBS). Leefstijl Nederlander Niet Verbeterd (Lifestyle Dutch Not Improved). *Press release PB07-021*, 2007. Retrieved February, 2008, from http://www.cbs.nl/NR/rdonlyres/EFCD6D79-C4A4-42C4-BCA9-D8EDFE67039E/0/pb07n021.pdf.

[41] Statistics Netherlands (CBS). Age and Overweight Prime Risk Factors for Diabetes. *Web Magazin*, 2007. Retrieved February, 2008, from http://www.cbs.nl/en-GB/menu/themas/gezondheid-welzijn/publicaties/artikelen/archief/2007/2007-2232-wm.htm.

[42] J.T. Tufano and B.T. Karras, Mobile eHealth Interventions for Obesity: A Timely Opportunity to Leverage Convergence Trends, *Journal of Medical Internet Research* **7**(5) (2005), e58.

[43] A. Stucki, P. Daansen, M. Fuessl, A. Cieza, E. Huber, R. Atkinson, N. Kostanjsek, G. Stucki and J. Ruof, ICF Core Sets for Obesity, *Journal of Rehabilitation Medicine* **44** (2004), 107–113.

[44] H.J.M. Vrijhoef, J. Janssen-Boyne, G. Engering, J.A. Kragten, G.J. De Weerd, M. Frederix and A.P.M. Gorgels, The health buddy: next term telemonitoring system for patients with heart failure, *European Journal of Heart Failure Supplements* **5**(1) (2008), 174.

[45] G.C. Williams, E.L. Deci and R.M. Ryan, Building Health-Care Partnerships by Supporting Autonomy: Promoting Maintained Behavior Change and Positive Health Outcomes, in: *Partnerships in Healthcare: Transforming Relational Process,* A.L. Suchman, P. Hinton-Walker and R. Botelho, eds, Rochester, NY: University of Rochester Press, 1998, pp. 67–87.

[46] World Medical Association (WMA). *World Medical Association Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Subjects,* 2004. Retrieved on February, 2008, from http://www.wma.net/e/policy/b3.htm.